

tests/regression/ test_result_returns_zero_test.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:**
Active standing regression guard (§11.4.135) for BUGFIX-0003.

Companion (§11.4.18) to the in-source documentation block at the top of the script. Pairs with docs/issues/fixed/BUGFIXES.md (BUGFIX-0003).

Overview

A standing regression guard that proves tests/run-tests.sh's test_result reporting helper **always returns 0**, even on a FAIL with no message. The pre-fix helper ended on a bare `[[ -n "$message" ]] && short-circuit` that returns 1 when the message is empty; when such a test_result was the last command of a test function, the function returned 1 and `set -euo pipefail` aborted the whole suite mid-run — most tests never executed and no summary printed (BUGFIX-0003).

It does **not** grep the source. GREEN extracts the **real** current test_result from tests/run-tests.sh, runs it under `set -euo pipefail` with a no-message FAIL, and asserts the process survives (exit 0).

Prerequisites

- bash (the extracted test_result uses `bash [[ ]] / local`; the guard wraps it in a bash probe so the outer `/bin/sh` parser stays POSIX-clean, §11.4.67).
- mktemp.

Usage examples

```
# GREEN guard (default) – assert the real test_result survives a  
# no-message FAIL:  
tests/regression/test_result_returns_zero_test.sh  
exit 0 = PASS
```

```
# RED reproduce – run the pre-fix replica and assert it aborts:
RED_MODE=1 tests/regression/test_result_returns_zero_test.sh #
    exit 0 = defect reproduced

# Runs automatically inside the suite:
bash tests/run-tests.sh #
    test_regression_guards()
```

§11.4.115 RED_MODE polarity

RED_MODE	What it runs	PASS means
0 (default)	the REAL test_result extracted from tests/run-tests.sh	it survives a no- message FAIL under set -e → the fix is present (GREEN guard)
1	a faithful PRE-FIX replica (last command is the bare &&)	it aborts under set -e → the defect reproduces

A RED_MODE=1 run that *cannot* reproduce (replica survives) is a finding per §11.4.7, not a pass.

Edge cases

- **Fix reverted** (the return 0 removed from test_result) → the GREEN guard reports
REGRESSION: test_result aborts the suite ... and exits 1. This is the §1.1 paired-mutation behaviour (proven byte-identical-restore in BUGFIX-0003).
- **Temp probe** is created with mktemp and removed on every exit path (trap ... EXIT INT TERM, §11.4.14).

Internal behaviour

- #!/bin/sh, set -eu; POSIX-only outer script (the bash probe is built into a temp file and run with bash).
- GREEN extracts test_result verbatim via awk from the tracked suite so it always tests the actual current code, never a stale copy.
- Writes one evidence file per run under qa-results/regression/bugfix0003/ (gitignored).

Related

- Fix site: `tests/run-tests.sh test_result()` (return 0).
- `docs/issues/fixed/BUGFIXES.md` — BUGFIX-0003 root-cause + verification.
- `tests/run-tests.sh` — registers this guard via `test_regression_guards()`.
- Sibling guard: `tests/regression/log_dir_writable_test.sh` (BUGFIX-0002).

Last verified

2026-07-01 — `sh -n + bash -n parse-clean`; RED reproduces (prefix replica aborts, `rc=1`), GREEN proves the fix (real `test_result` survives); §1.1 mutation makes the GREEN guard FAIL and `tests/run-tests.sh` restores byte-identical (md5 `7c2bab18c4566d081b1c8aa7a9a412e0`). Full suite runs to completion (41 tests, summary printed).